

Pseudocode: Employee Attendance and Overtime Analyser

START

Load employee records from the dataset.

Set valid_records to an empty list.

Set invalid_records to an empty list.

FOR each employee record:

- Validate that the employee ID and name are present.
- Validate that the department is present.
- Validate that expected days are greater than zero.
- Validate that present days are not negative.
- Validate that leave days are not negative.
- Validate that present days plus leave days do not exceed expected days.
- Validate that hours worked are not negative.
- Validate that the hourly rate is greater than zero.
- Validate that employment type is recognised.

IF any validation fails:

- Store the employee record in `invalid\_records`.
- Store the reason or reasons for rejecting the record.

ELSE:

- Store the employee record in `valid\_records`.

END IF

END FOR

FOR each employee record in valid_records:

- Calculate the number of absence days by subtracting present days and approved leave days from the expected days (Present+Approved-Expected)
- Calculate the adjusted attendance percentage.

IF attendance percentage is at least 95%:

Classify attendance as "Excellent".

ELSE IF attendance percentage is at least 85%:

Classify attendance as "Satisfactory".

ELSE:

Classify attendance as "Needs improvement".

END IF

Determine standard monthly hours
according to employment type:

IF employment type is Full-time:

Set standard hours to 160.

ELSE IF employment type is Part-time:

Set standard hours to 80.

ELSE IF employment type is Contract:

Set standard hours to 120.

END IF

Calculate overtime hours.

IF overtime hours are negative:

Set overtime hours to zero.

END IF

Calculate regular payment.

Calculate overtime payment
using 1.5 times the hourly rate.

Calculate total payment
by adding regular payment and overtime payment.

END FOR

Display the main menu.

WHILE True:

Display menu options.

Get the user's choice.

IF choice is 1:

Display all valid employee records.

ELSE IF choice is 2:

Ask for an employee ID or name
and search the valid employee records.

Display the matching employee
or an appropriate "not found" message.

ELSE IF choice is 3:

Analyse and display employee attendance,
including attendance percentage,
absence days and attendance category.

Identify employees with the highest
and lowest attendance.

ELSE IF choice is 4:

Display employees whose attendance
is classified as "Needs improvement".

ELSE IF choice is 5:

Analyse and display overtime information.

Identify employees with:

- the most overtime hours;
- the highest overtime payment;
- the highest total payment.

ELSE IF choice is 6:

Calculate and display each department's:

- average attendance;
- overtime expenditure.

Identify the department with
the highest overtime expenditure.

ELSE IF choice is 7:

Display all invalid employee records
together with their rejection reasons.

ELSE IF choice is 8:

Calculate and display the organisational summary,
including:

- total supplied records;

- valid records;
- invalid records;
- total overtime hours;
- average overtime hours;
- total overtime expenditure;
- average attendance;
- attendance category counts;
- employees with unexplained absences.

ELSE IF choice is 9:

Display the program exit message.

BREAK the loop.

ELSE:

Display "Invalid selection."

Return to the main menu.

END IF

END WHILE

END